



Full Length Article

Experimental and numerical study of radial and axial HTS magnetic couplers

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ABSTRACT

The paper presents the results of an experimental study and numerical calculations of torques in magnetic couplers of various configurations. A comparison was made for model couplers based on permanent magnets (PMG) and couplers based on high-temperature superconducting (HTS) tapes cooled with liquid nitrogen. In PMG-couplers, both coupler halves are made of permanent magnets, while in HTS-couplers, one coupler half is made of HTS-tapes stacks, and the second of permanent magnets. Both types of couplers are considered in both axial and radial configurations, with different clearances between the coupler halves. The results of experimental studies showed that in the configurations we considered, radial HTS-couplers are superior to their axial analogues in terms of the maximum torque between the coupler halves. In addition, in the considered configurations, PMG couplers exceeded their HTS analogues in torque by 10 times. Numerical finite element calculations using a combined A-T-H formulation were used to optimize the design of the HTS-coupler. The model was verified using the obtained experimental data for moments in model couplers. The model was used to calculate an optimized coupler with added HTS-windings, lower HTS temperature and stronger permanent magnets. The calculation results showed that HTS-couplers can be comparable with their PMG analogues, at the same time they are significantly superior to them in damping external disturbances. Due to the fact that the best characteristics were shown by model radial couplers with minimal gaps between the coupler halves, radial couplers were manufactured based on windings of 10 and 20 layers of HTS-tapes and the torques in them were measured in the FC and ZFC modes. It has been shown that in most operating conditions, 10-layer couplers outperform 20-layer couplers.

1. Introduction

One of the important elements of a flywheel energy storage device is a magnetic coupler. It allows a mechanical connection between the flywheel and the motor/generator. When the clutch connects the engine to the flywheel, it allows it to accelerate, thereby increasing the flywheel's stored kinetic energy. When the flywheel is connected to the generator, rotational energy is collected and converted into electrical energy. But most of the time the kinetic storage device operates in energy conservation mode. In this mode, the coupler must ensure that there is no interaction between the flywheel and the motor/generator to minimize energy dissipation. Typically, flywheel energy storage devices use a universal motor-generator that can operate in both energy storage and energy withdrawal modes. Then in such systems it is enough to use one coupler.

In flywheel energy storage devices, it is advisable to place the flywheel in a vacuum volume, which eliminates the friction force and, consequently, energy losses in the energy conservation mode [1,2],

and also to use superconducting magnetic bearings. Stability and almost complete absence of friction even under heavy loads makes them almost irreplaceable in flywheel energy storage systems. It is on the development of kinetic energy storage that work in the field of superconducting bearings has been focused in the last decades [3,4]. However, standard motors/generators are designed for use under normal conditions. Therefore, it is advisable to use non-contact couplers for superconducting energy storage devices. The most common are magnetic couplers, which allow communication between the flywheel and the engine/generator through magnetic interaction. Magnetic couplers are made on the basis of permanent magnets and have been known for a long time [5,6]. These systems are actively used in devices where contactless transmission of motion is necessary: water pumps, underwater turbines, etc. [7]. Permanent magnet couplers can achieve high torque limits, but require mechanical retraction of one part of the coupler from the other to operate in energy storage mode. Also, a big disadvantage of such systems is their instability, due to the fact that the system is in operating condition at maximum

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potential energy, and when a shift occurs from this position (for example, due to parasitic vibrations), the system becomes unstable.

An alternative to permanent magnet couplers can be couplers based on high-temperature superconductors (HTS). Since the concept of an HTS flywheel energy storage device obviously assumes the presence of a cryogenic cooling system, the integration of a superconducting magnetic coupler into the device does not require significant design changes. A flywheel storage device's superconducting bearing cooling system can be used to cool superconductors. An important advantage of superconducting couplers is that they have the property of self-stabilization and, unlike permanent magnet clutches, do not require additional radial and axial stabilization of the flywheel. Another advantage of superconducting couplers is that the coupler can be opened and closed by heating and cooling the superconductor, without the additional axial forces that appear in permanent magnet couplers. Another important advantage of superconducting couplers is the hysteresis of the interaction force between the parts of the coupler during their mutual movement. This leads to rapid attenuation of parasitic vibrations caused by various external influences. These advantages make superconducting couplers a promising element of flywheel energy storage devices.

The most promising high-temperature superconducting materials are HTS-coated conductors tapes (CC-tapes). They make it easy to create superconducting assemblies of various configurations; the presence of a metal substrate makes the material durable and able to withstand loads caused by volumetric forces in strong magnetic fields, and copper stabilizing layers allow efficient and uniform cooling of the superconductor and avoid its overheating in alternating magnetic fields. There are a number of works on the study of superconducting couplers [8–10], but all are focused on the use of CC-tapes windings as an element of the coupler.

In general, a magnetic coupler is a device consisting of two coupler halves and acting as a non-contact transmission mechanism. The driving coupler half is driven from the outside and transmits torque to the driven coupler half. The driven coupler half rotates the shaft with a massive flywheel. The key characteristic of the coupler is the maximum clutch torque, which determines the power of energy removal or supply N , defined as the product of the modulus of the moment of force M by the rotational speed ω :

$$N = M \cdot \omega \quad (1)$$

The required rotation speed of the coupler depends on the amount of energy that needs to be supplied or removed. In the vast majority of developed designs of HTS flywheel energy storage devices, the gravity force of a massive flywheel is compensated using internal bearings and suspensions, so in order to spin up the shaft it remains necessary to overcome the moment of inertia of the driven coupler half, which is connected to the flywheel shaft. Obviously, the critical parameters of the coupler are the maximum torque, which the coupler can withstand without turning, as well as rotational stability.

2. Preliminary tests of various couplers configurations

2.1. The coupler designs

There are two main types of magnetic couplers – radial and axial [6]. In our work, both of these types are considered for two versions – based on permanent magnets (PMG-PMG) and based on superconducting tapes (HTS-PMG). Schematic images of the couplers considered in our work are presented in Fig. 1.

In the radial geometry of the magnetic coupler (Fig. 1a), the magnetic systems of the coupler halves are the outer and inner rotors. An

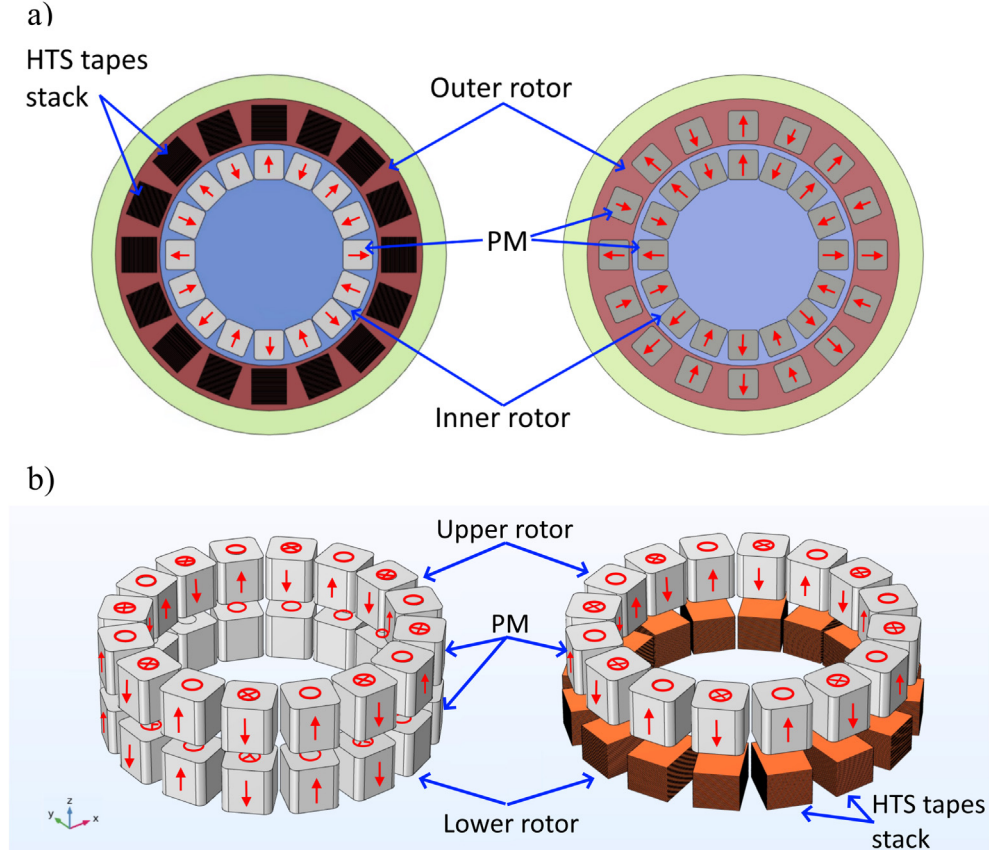


Fig. 1. A) schematic representations of radial magnetic couplers. left – hts-pmg coupler, right – pmg-pmg coupler. b) schematic representations of axial magnetic couplers. on the left is pmg-pmg coupler, on the right is hts-pmg coupler.

assembly of sixteen cubic PMGs with a side of 10 mm was chosen as the basis for the driving coupler half (inner rotor), as was used in the development of the HTS bearing in our previous work [11]. Permanent magnets of these sizes were chosen because they are commensurate with the width of the HTS tape and are easy to purchase. Permanent magnets are located on tangents to a circle of diameter of 51 mm. This PM assembly was manufactured for experimental studies and was used to compare the force characteristics of PMG-PMG couplers and HTS-PMG couplers.

The driven rotor of a radial HTS-PMG coupler (Fig. 1a, left) consists of sixteen stacks of 12x12mm CC-tapes, with the geometric center of each stack and the center of each magnet located at the same radius of the coupler circumference. The results of test calculations showed that the use of an assembly of co-directional PMs in the driving half-clutch at speeds of more than 10 Hz and a flywheel weight of more than 1 kg can lead to slippage of the HTS-PMG coupler. In order to avoid this, it is necessary that the current in neighboring stacks of CC-tapes be induced in opposite directions. That is, the magnetic moments of neighboring PMGs must be counter-oriented.

Schematic diagrams of axial magnetic couplers are presented in Fig. 1b. Similar to the case of a radial coupler, a block of 16 permanent magnets with dimensions of 10x10x10 mm is selected as the basic element of the magnetic system. By default, the upper coupler half is the driving one. In the case of a PMG coupler (Fig. 1b, left), the lower driven coupler half completely duplicates the driving half. The magnetic moments of the PM are directed along the axis of rotation, and neighboring magnets are magnetized in the opposite direction. The driven coupler half in the case of a HTS-PMG coupler (Fig. 1b, right) is assembled from 16 stacks of CC-tapes, assembled from pieces of 12x12mm tapes.

2.2. Experimental methods

To study the load characteristics of couplers of various configurations, the dependence of the torque between the coupler halves on the angle of their mutual rotation was measured. To simplify the experiment, the driven coupler half was replaced by one of its elements: one stack of HTS tapes for the superconducting coupler or one magnet for the PM coupler. Fig. 2 shows schematic images of the experimental geometry for couplers of radial and axial configuration. Due to axial symmetry, to measure torque, it is sufficient to use a driving half-coupler with permanent magnets (1 in Fig. 2a and 2b), consisting of 16 elements, and a stationary part consisting of one element – one stack of HTS tapes or one permanent magnet (2 in Fig. 2a and 2b). Then the resulting torque will be equal to 1/16 of the torque arising in the coupler, consisting of 2 full parts of 16 elements, and the total torque in the final coupler will be 16 times greater than in model couplers. However, for a comparative analysis of different types of couplers, measurements on model couplers will be sufficient. The moment of force was calculated as the vector product of the interaction force F , which is directly measured using a load cell, and the force arm L :

$$M = |\vec{M}| = |\vec{L} \times \vec{F}| = F \cdot L \quad (2)$$

However, it should be noted that this simplification of the experiment introduces some distortion into the measured torque. A stack of HTS tapes introduces distortion into the picture of magnetic field distribution. When using a full-fledged coupler, each stack of HTS tapes will be in a magnetic field distorted by other stacks, which can change the dependence of the torque on the angle. Therefore, in section 2.4 of the article we presented a comparison of experimental data measured by a simplified method with the results of modeling a full-fledged system of 16 stacks of HTS tapes. Good agreement between calculation and experiment demonstrated the validity of the simplified experimental approach.

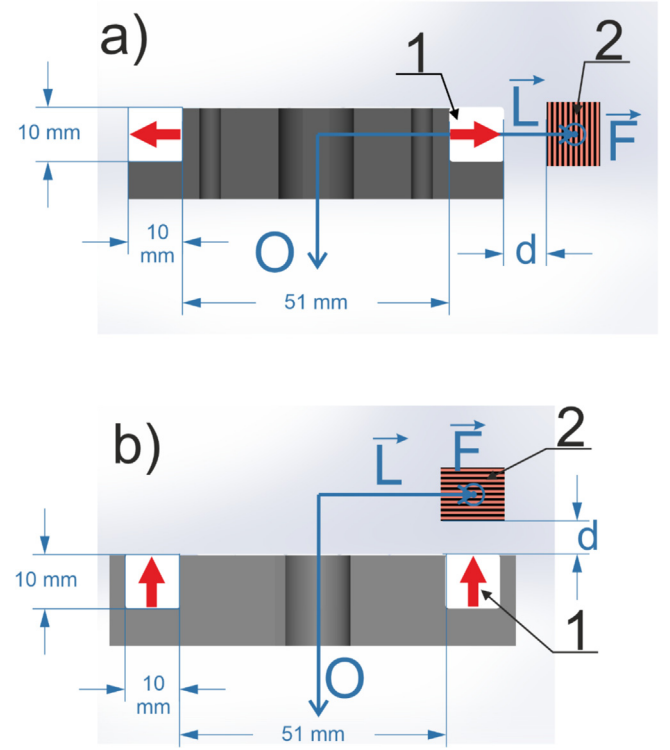


Fig. 2. Schematic representation of the relative position of the coupler parts in the study of the torque in the radial coupler (a) and the axial coupler (b). 1 – coupler with permanent magnets, 2 – stack of HTS tapes or permanent magnet. The rotation angle of the coupler and the force F were measured. d is the distance between the two parts of the coupler. The red arrows indicate the magnetization of permanent magnets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To study the interaction of couplers, we used a setup whose diagram and photograph is shown in Fig. 3. An example is given for radial geometry. The PMG coupler half (1 in Fig. 3) is attached to a stepper motor with a gearbox (2 in Fig. 3). The minimum angular rotation step is 0.045° . The element of the driven coupler half (3 in Fig. 3) is attached to a load cell (4 in Fig. 3) to measure the interaction force. In the case of studying of HTS-PMG coupler, the element of the driven coupler halves (a stack of HTS tapes) is placed in a cryostat with liquid nitrogen (5 in Fig. 3). To position the element of the driven coupler half, a 3-coordinate positioning system was used (6 in Fig. 3).

During the experiment, the tangential component of the interaction force between the coupler half element fixed in the holder and the driving half-coupler with permanent magnets was measured using a strain gauge. This force was measured for various rotation angles of the drive half, thus giving the interaction force as a function of angle.

For the manufacture of driving coupler's half and as an element of the driven half of PMG-PMG coupler, cubic N42 NdFeB permanent magnets with a side of 10 mm were used. The residual induction on the surface of each permanent magnet was 0.1 T.

The element of the HTS-PMG coupler was stack of 50 layers of HTS tapes placed in a cryostat and cooled with liquid nitrogen (3 in Fig. 3). We used tapes produced by S-Innovation [12], 12 mm wide, with copper coating. The tape architecture is based on cold rolled Hastelloy C276 substrate, biaxially textured MgO buffer layer deposited by ion beam assisted deposition, and $\text{YBa}_2\text{Cu}_3\text{O}_7$ HTS layer deposited by pulsed laser deposition. The total thickness of the tape is 120 μm . The average critical current of the tape, measured in self-field in liquid nitrogen, is about 550 A. The tape was cut by use of guillotine paper cutters into fragments with sides of 12x12 mm² and assembled into

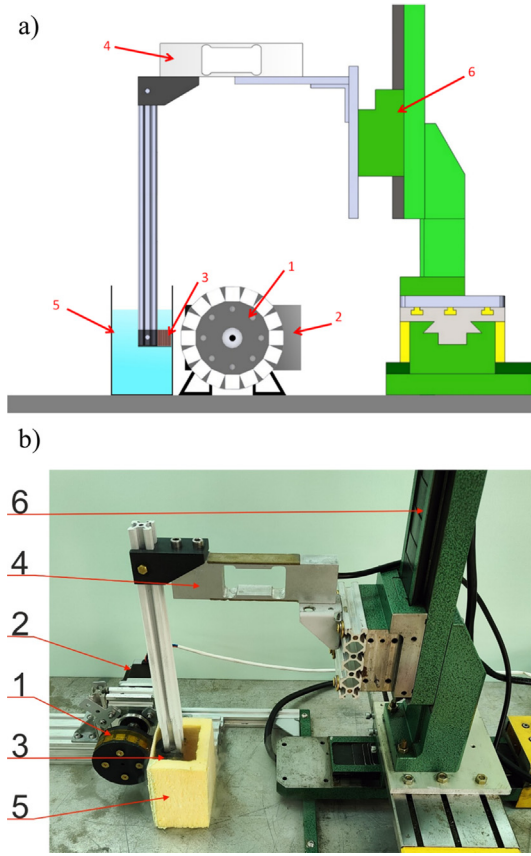


Fig. 3. Schematic diagram (a) and photograph (b) of the experimental setup for measuring the interaction forces of coupler elements. 1 – driving half-coupler with PMG, 2 – stepper motor with gearbox, 3 – element of the driven half-coupler, 4 – load cell, 5 – cryostat with liquid nitrogen, 6 – positioning system.

a stack of 50 such fragments. The final thickness of a stack of 50 layers is 6 mm. Stacks of 50 layers were used to increase the values of the magnetic interaction force, which makes it possible to reduce the error in the experimental measurement of the interaction force using a load cell. In our experience, for tapes with a critical current of 550A when interacting with most commercially available permanent magnets [CR curr], the optimal ratio of levitation force to the number of layers in a stack is 20 layers. With a further increase in the number of elements in the stack, the increase in levitation force slows down. 50 layers were used in a stack, as a balance between the force of interaction and the material consumed.

2.3. Experimental results

As a result of measurements, we obtained dependences of the force acting on a single stack/magnet on the rotation angle of the coupler with permanent magnets. The obtained data were recalculated into the moment of force as the product of the measured force F by the lever L (in accordance with formula 1).

The obtained dependencies for axial and radial PMG-PMG couplers are shown in Fig. 4. Measurements were carried out for 3 distances between the parts of the coupler (d in Fig. 2): 5 mm, 6 mm and 7 mm. When shifted by an angle of 22.5 degrees (1/16 of a full rotation), the outer magnet is opposite the oppositely directed magnet, the system force returns to zero. After rotation for another 22.5 degrees, the system returns to a position similar to the original one. It follows from the dependencies that the maximum force at different rotation cycles changes slightly – within 5 %. The changes in force are

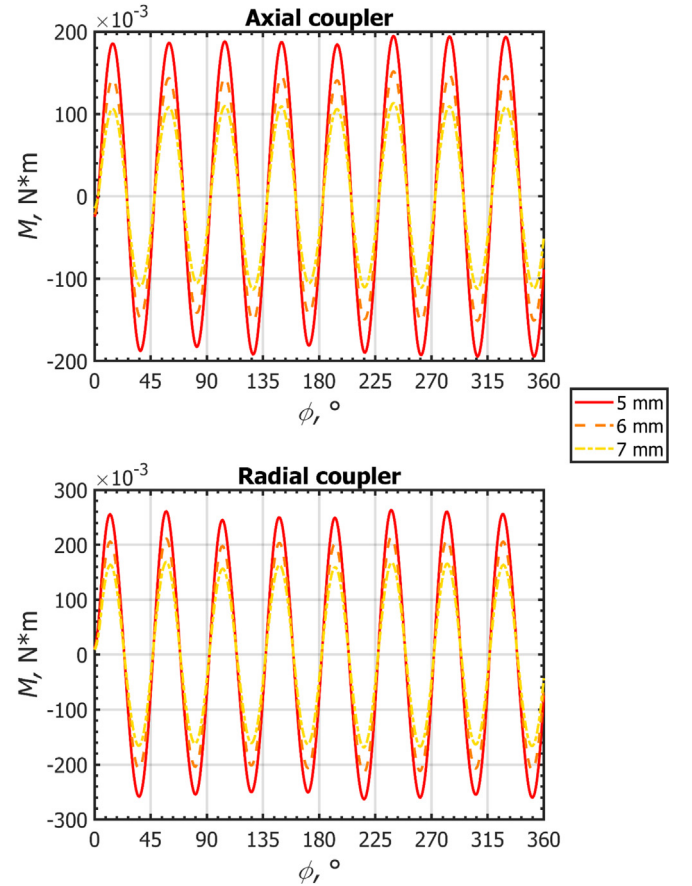


Fig. 4. Dependence of the torques between the parts of the coupler on the rotation angle for an axial and radial PMG-PMG couplers. Results are presented for different distances between coupler halves: 5 mm, 6 mm and 7 mm.

due to slight differences in the magnetization of the individual permanent magnets and in their position relative to the stationary half coupler. In general, for all types of permanent magnet couplers, a periodic torque dependence is obtained with a period of 45 degrees.

But this picture is distorted for the case of HTS-PMG couplers, the dependences of the moment on the angle of rotation for which are shown in Fig. 5. From the dependencies it is clear that the maximum force varies from cycle to cycle, which is primarily associated with the processes of remagnetization of HTS-tape stacks.

For a radial HTS-PMG coupler, with a distance of 5 mm, the maximum force is observed in the first cycle, after which it drops by ~ 20 %, and further changes do not exceed 5 %. At large distances between the coupler parts, the force in the first cycle does not prevail over the remaining cycles. One can see a slight difference in the dependence on the sinusoid near zero torque, which is more noticeable at large distances.

For an axial coupler, the maximum force in the first cycle is higher than in subsequent cycles for any distance between the coupler parts. We also note that at a distance of 5 mm, the remagnetization of the HTS occurs near zero torque, which leads to the appearance of additional peaks. This is undesirable for superconducting couplers because leads to instability of the system near the equilibrium position. This makes axial HTS-PMG couplers HTSC less preferable for use as an element of a kinetic storage device.

For convenience of comparison of the maximum torques, Fig. 6 shows the dependences of the maximum torque on the distance between the coupler elements for a superconducting coupler (HTS-

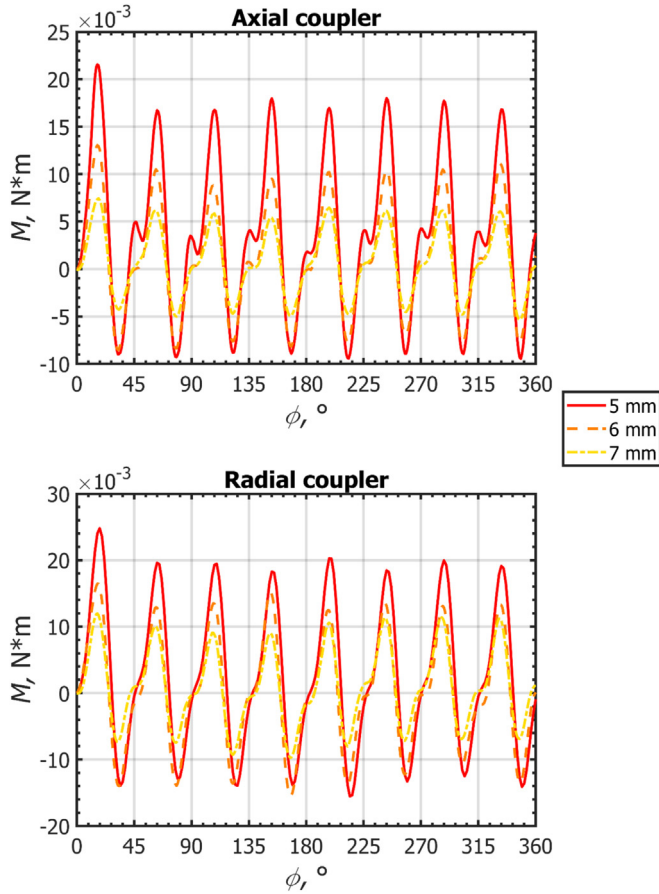


Fig. 5. Dependence of the torques between the parts of the coupler on the rotation angle for an axial and radial HTS-PMG couplers. Results are presented for different distances between coupler halves: 5 mm, 6 mm and 7 mm.

PMG) and a coupler based only on permanent magnets (PMG-PMG) in radial and axial configurations.

It follows from the dependencies that the torque in both types of coupler increases with decreasing distance between the parts of the coupler. For HTS-PMG couplers, the moment in the radial coupler noticeably exceeds the moment in the axial coupler. At a distance of 5 mm, the force gain is about 15 %, while for a distance of 7 mm the force gain is 65 %. For PMG-PMG couplers, the differences in torque between radial and axial couplers are not as significant as in the case of HTS-PMG couplers. The relative increase in torque with increasing distance is not so noticeable and varies from 15 % for 5 mm to 23 % for 7 mm.

Another fundamental difference between HTS-PMG and PMG-PMG couplers is the non-zero average value. In PMG-PMG couplers, the torque fluctuates around the zero value – the force graph turns out to be antisymmetric with respect to the zero coordinates. In the case of HTS-PMG couplers, oscillations occur around a positive average value. The presence of non-zero torque has a strong effect on the performance of the coupler. If the rotation speeds of the coupler's halves are not synchronized (that is, the coupler's halves slip relative to each other), then in the case of PMG-PMG couplings the average torque will be zero. Then, in accordance with formula 1, the power of energy removal or supply N will be equal to zero. In the case of HTS-PMG, the average torque is not equal to zero. Then, even in the case of slippage, there will be non-zero power of energy removal or supply N .

Data on the average value and amplitude of oscillations for HTS-PMG couplers are shown in Fig. 7. It can be seen that the average torque value is significantly lower than the amplitude of the torque oscil-

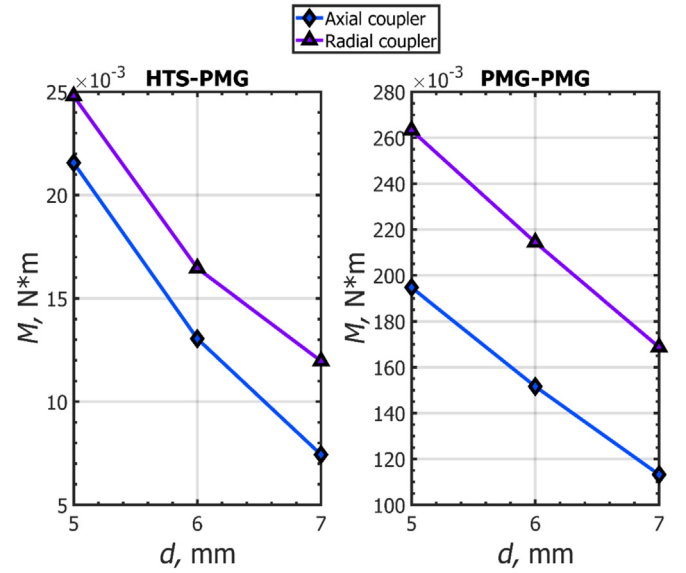


Fig. 6. Dependence of the maximum torque on the distance between the half couplers. Results are presented for a) a superconducting coupler (HTS-PMG) and b) a permanent magnet-only coupler (PMG-PMG) in radial and axial configurations.

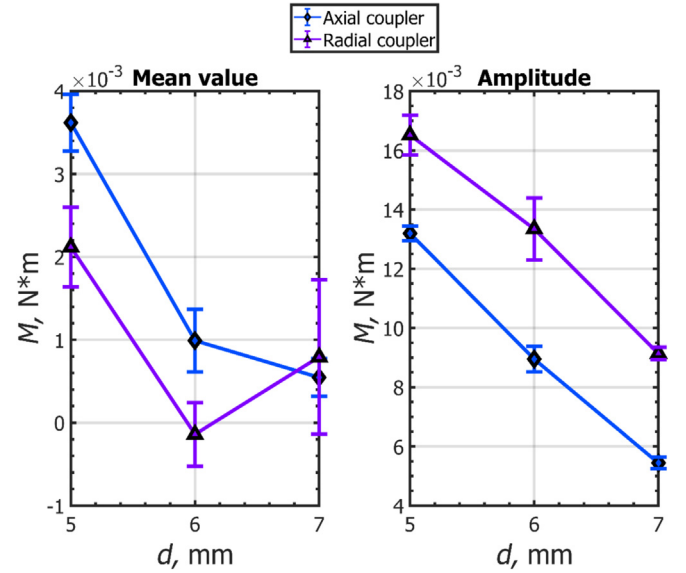


Fig. 7. Dependences of the average value and amplitude of the torque in HTS-PMG couplers on the distance between the coupler halves. Results are presented for couplers in radial and axial configurations.

lations. Both the amplitude and the moment of force decrease with increasing distance between the coupler halves. At the same time, the shift of the average value is higher for axial couplers for distances of 5 and 6 mm. For a distance between half couplers of 7 mm, average torque value is less than the measurement error, which makes it difficult to assess the relationship between the average torque values of two types of couplers.

One more important difference between PMG and HTS couplers is that in PMG couplers the interaction force is significantly higher, approximately 10 times. The experiment used stacks of 50 HTS tapes. Taking into account the nonlinear growth of the interaction force with the thickness of the stack, with stacks of about 100 tapes, one can

expect an increase in force by 20 % [13,14], and when using cryocoolers for cooling to temperatures of the order of 40 K, which is used in real industrial systems, the force can increase another 1.5–2 times [15]. Then couplers based on HTS can be comparable in torque to couplers based on permanent magnets. However, for a more accurate comparison, an experimental study or simulation of the optimized coupler is necessary. The simulation results will be presented in the next section.

2.4. Numerical simulation

Numerical modeling of radial (2D) and axial (3D) magnetic couplers was performed using the finite element method in the engineering simulation environment Comsol Multiphysics. A combined A-T-H formulation was used. A detailed description of the numerical model can be found in [11]. Here we provide a brief description of the main elements of the model.

For modeling, the Rotating Machinery, Magnetic (RMM) module was used, combining formulations in terms of magnetic vector potential (MVP) and magnetic scalar potential (MSP). The use of a mixed formulation can significantly reduce computation time, and also allows the use of the moving mesh principle when modeling rotating machines in three-dimensional geometry.

The formulation in terms of scalar potential is only applicable to regions in which the electric current density is zero. In this case, a scalar field V_m (magnetic scalar potential, MSP) is introduced and the magnetic field is defined as the gradient from this potential. With this definition, Ampere's law is satisfied automatically and the law of conservation of the magnetic induction vector is solved. Since in all the considered designs of magnetic couplers, the driven coupler half is an assembly of 16 permanent magnets, that is, an area with zero current, the MSP formulation applies specifically to the driven coupler halves and the domains adjacent to them.

To calculate the driving coupler half, a vector field A (magnetic vector potential), is introduced, and the magnetic induction and electric field vectors are determined as $E = -\frac{\partial A}{\partial t}$ and $B = \nabla \times A$. As a result of this definition, the fields B and E automatically satisfy two of Maxwell's equations: Faraday's law and the law of conservation of magnetic induction (or Gauss's law for the magnetic field). This can be represented as $\nabla \times E = -\frac{\partial B}{\partial t}$ and $\nabla \cdot B = 0$. The equation to be solved is Ampere's law $\nabla \times H = J$.

For modeling, the dependence of the critical current lift factor on temperature and magnetic field for HTS samples is introduced as an interpolation of experimental data.

In the simulation, two magnetization modes were considered. In the first mode, the HTS elements are placed in the PMG field and then cooled to a given temperature with a non-zero initial value of the external field (FC mode). In the second mode, pre-cooled stacks of HTS tapes are placed opposite permanent magnets with zero initial magnetization. Then, the magnetization value of the PMG is linearly increased to 0.1 T in 10 s. This simulates the zero-field cooling mode (ZFC). After this, the system is kept stationary for another 10 s to ensure complete stabilization of thermal processes after magnetization. Then rotation begins. In both modes, as part of the model experiment, the drive coupler half rotates at a frequency of 2 Hz with a force of 500 N. With such a force, both HTS couplers inevitably slip, which makes it possible to study the force and torque of their interaction at different displacement angles. Magnetization of stacks of HTS tapes occurs at a temperature of 77.4 K. Fig. 8 shows a comparison of the calculated and experimental dependences of the moments, normalized per one magnetic element (1 PM out of 16) on the angle of rotation of the PM-PM coupler. The results are presented for a radial coupler with a gap between the coupler halves of 5 mm. From a comparison of the dependences, it is clear that the obtained calculated curves are in good agreement with the experimental data. This confirms the

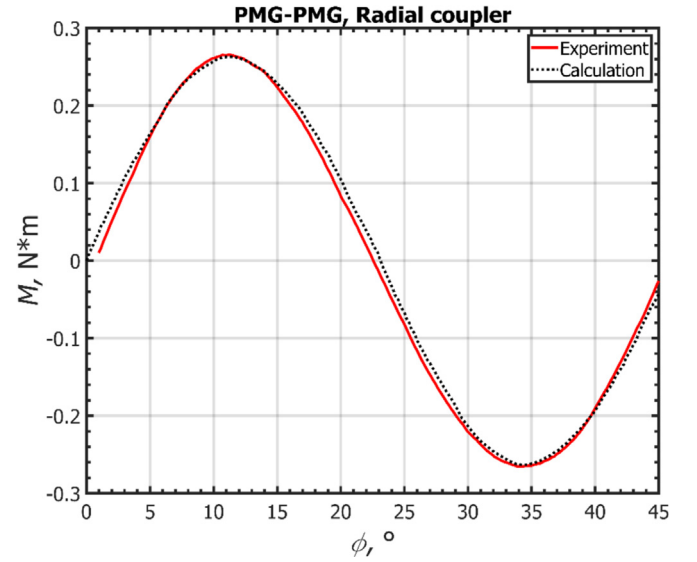


Fig. 8. Calculated and experimental dependences of the torque on the rotation angle of the radial PMG-PMG coupler, normalized to one permanent magnet. The distance between the coupler halves is 5 mm.

correctness of the calculation model and allows it to be used to predict the behavior of permanent magnet couplers. The model also performed well for axial coupler calculations.

For the case of HTS-PMG couplers, the interaction was calculated in the case of cooling with liquid nitrogen. A comparison of experimental and calculated curves for a radial coupler with a distance between the parts of the coupler is shown in Fig. 9. Similar to the case of PMG-PMG couplers, the values of the torques are given, normalized per one magnetic element (1 stack of HTS tapes out of 16). It can be seen that the calculated dependence well describes the experiment in the first half-period of the curve. In this case, the maximum moment of interaction force is determined quite accurately. However, in the second half of the half-cycle, where a stack of HTS tapes falls into a field of the oppo-

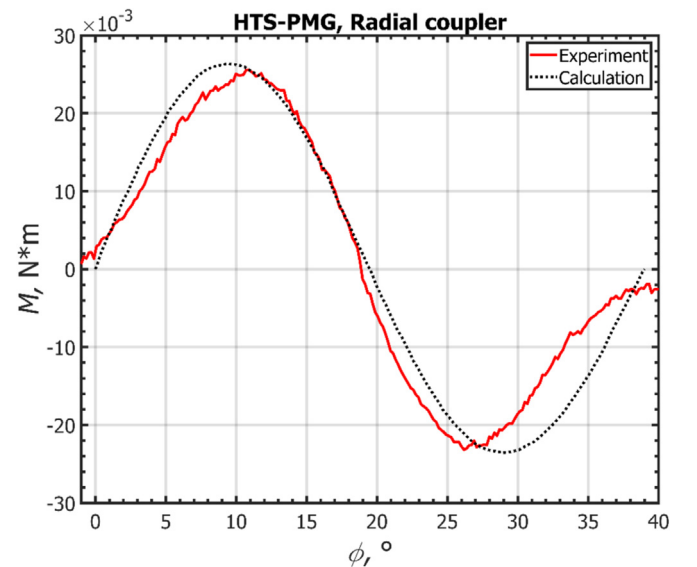


Fig. 9. Calculated and experimental dependences of the torque on the rotation angle of the radial HTS-PMG coupler, normalized to one permanent magnet. The distance between the coupler halves is 5 mm. Temperature 77.4 K.

site sign, the calculated force converges worse with the experiment. This can be explained by the strong influence of tape remagnetization on the coupler torque.

Thus, the proposed model well predicts the maximum torque for PMG-PMG and HTS-PMG couplers, and can be used for the development of superconducting couplers. The model was used to optimize the performance of a superconducting coupler. Based on the obvious lack of strength and rigidity of the HTS-PMG coupler, it is recommended to increase the number of HTS elements by adding HTS windings to the design of the coupler with the number of layers equal to 50. In addition, to achieve better performance on the interaction force, the temperature in the system should be lowered to 65 K, which will lead to an increase in the levitation force. In practice, such a temperature is easily achieved by pumping out nitrogen vapor, so the proposed procedure does not require the introduction of cryocooler technology into the working space of the coupler. Another step to increase the torque is to use stronger permanent magnets. For the considered cubic NdFeB magnets with a side of 1 cm, the maximum residual magnetic field induction on the surface is about 0.35 T. The final schematic diagram of the HTS-PMG coupler is shown in Fig. 10. The calculated load curves for the optimized coupler design in comparison with a PMG-PMG coupler of a similar configuration are presented in Fig. 11. From the comparison of the dependencies, it is clear that the maximum torque of the optimized HTS-PMG coupler is comparable in order of magnitude with the maximum moment in the PMG-PMG coupler. The average torque value of the optimized configuration is about 20×10^{-3} N/m, which significantly exceeds this value for the experimentally studied couplers (see Fig. 7 for comparison). This shows that even if the coupler's halves slip, energy will be removed/supplied to the flywheel, but the power removed/supplied will be approximately an order of magnitude lower.

The resulting design is practically not inferior to its PMG-PMG counterpart in terms of torque. Another advantage of the HTS-PMG coupler is its ability to damper external disturbances. Fig. 12 shows the dynamics of changes in the amplitude of movement of the couplers when a vertical exciting action with a pressure of 100 N/m^2 is applied to them. It is clearly seen that vibration damping in the case of a HTS-PMG coupler occurs much faster than in a design based on permanent magnets.

3. Research of torque in a radial coupler

3.1. The coupler design and experimental methods

After preliminary studies, it was decided to carry out measurements on a full prototype of a coupler based on HTS tapes. For a complete

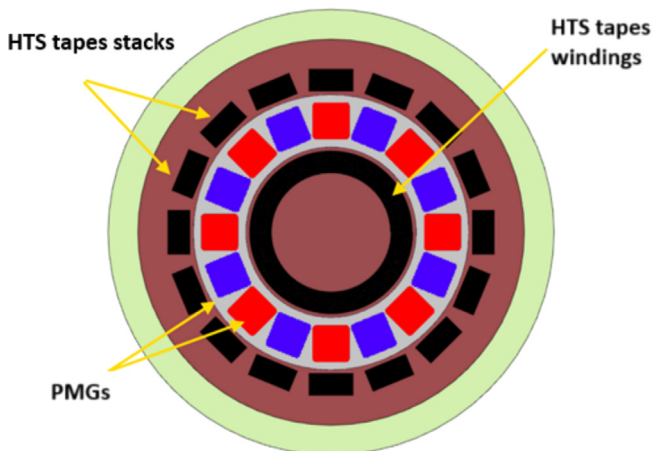


Fig. 10. Schematic diagram of an optimized HTS-PMG coupler.

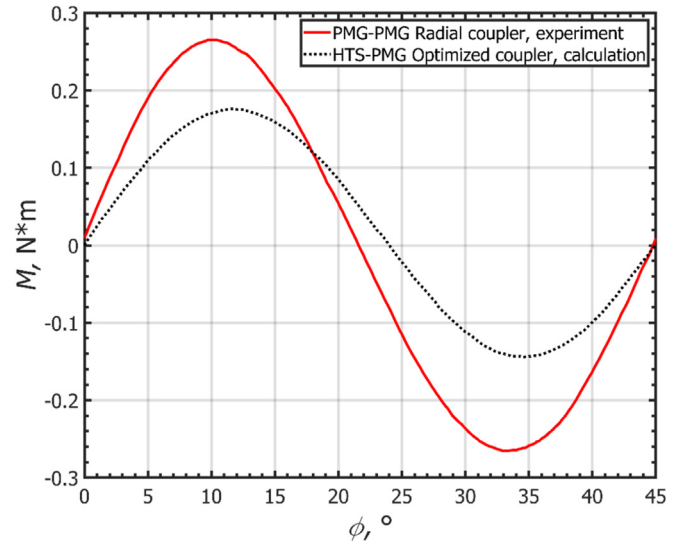


Fig. 11. Dependence of the torque on the rotation angle of the PMG-PMG radial coupler and the optimized HTS-PMG coupler. Temperature 65 K. Значения нормированы на 1 PMG.

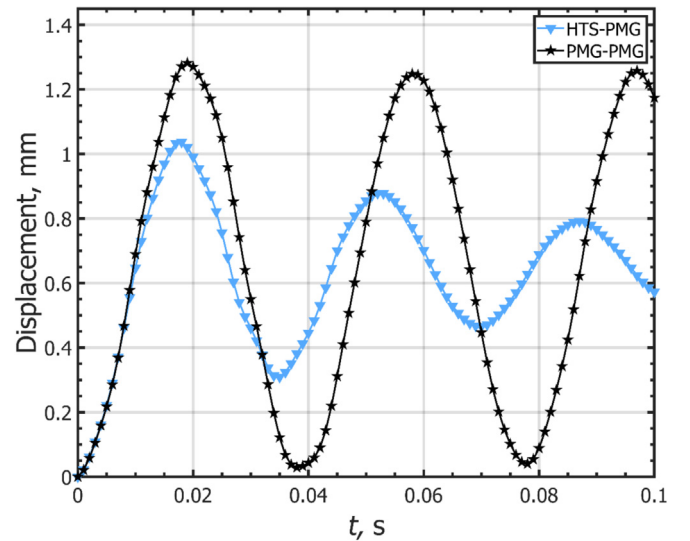


Fig. 12. Dependences of the vibration amplitude of HTS-PMG and PMG-PMG radial couplers on time when an instantaneous excitation with a pressure of 100 N/m^2 is applied to the couplers.

comparison of the two types of couplers (radial and axial), it is necessary to optimize each type of coupler and compare the optimized designs, which is beyond the scope of this work. But based on the research conducted in the previous section, it can be concluded that radial couplings are superior in maximum torque and torque stability to axial couplings in the configurations we considered.

For this reason, we decided to take as a basis the configuration corresponding to the radial coupler. Another reason for using a radial coupler is the ease of its manufacture in the laboratory. Stacks of HTS tapes in this coupler configuration can be replaced by one winding of HTS tape. To create stacks, one need to cut a large number of pieces of tape with high precision and assemble them into stacks. When cutting, the tape may degrade at the cut points. In addition, the superconductor at the cut site becomes accessible to the external environment. If it is not further insulated (for example, by soldering), the superconducting layer will begin to degrade over time. To create coils, the sim-

plest winding machine or manual winding is sufficient. Therefore, under laboratory conditions, experimental winding samples are usually much easier and faster to make. However, at the scale of commercial production, the difference in manufacturing complexity becomes insignificant.

It should also be noted that the manufacture of a radial coupler in the winding-based configuration we discussed earlier is impossible. This is due to the fact that in the configuration we considered, the plane of the HTS tape is located perpendicular to the axis of axial symmetry. And the creation of windings is possible only in the form of a cylindrical surface. And the use of windings in an axial configuration will require a significant change in the configuration of permanent magnets in order to create an optimal configuration of the magnetic field in the area of HTS tape. Therefore, based on the results of preliminary studies of couplers of various configurations, as well as taking into account the ease of manufacture in laboratory conditions, it was decided to carry out measurements of torque in radial couplers based on HTS-tape windings.

For the manufacture of windings, the same HTS tape produced by S-Innovation [12] was used (thickness 12 mm, copper coating, critical current – 550 A in self-field) as for the manufacture of stacks of HTS tapes. The windings are open looped coils. To create the HTS windings, the tape was wound onto a base, 3D printed from Polylactic Acid (PLA) plastic. The base is a cylinder with a closed bottom side, on the outer part of which a tape is wound. The tape was fixed during winding using copper adhesive tape and coated with glue between the layers of tape for better fixation. The photograph of the base with windings is shown in Fig. 13.

The schematic diagram of the experimental setup is shown in Fig. 14, the appearance of the experimental setup is shown in Fig. 15. The HTS-tape (1 in Figs. 14 and 15) is wound on a base (2 in Figs. 14 and 15) fixed in a cryostat made of polystyrene foam (3 in Figs. 14 and 15). Liquid nitrogen is used to cool the superconductor (4 in Figs. 14 and 15). The tape acts as a driven coupler half. The driving half-coupler (5 in Figs. 14 and 15) is the same half-coupler with 16 permanent magnets as for preliminary studies of radial couplers (Fig. 3.a). The driving half-coupler is located inside the winding base so that the centers of the permanent magnets are located opposite the center of the HTS tape. Due to the fact that the base is closed from bot-

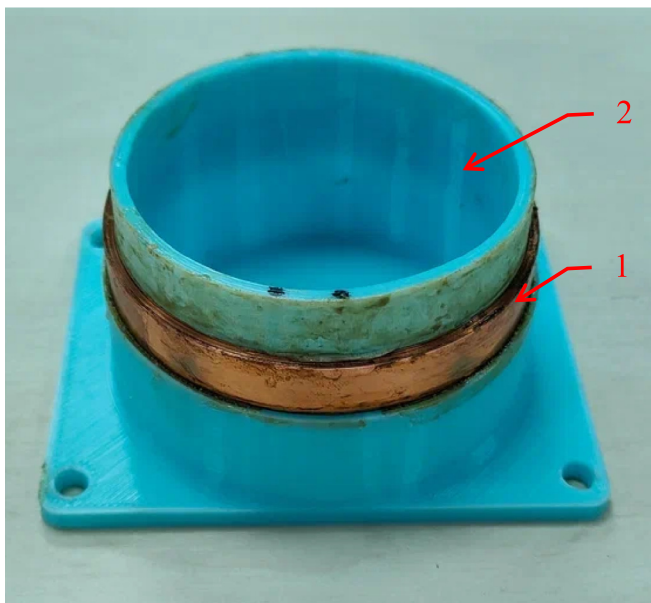


Fig. 13. Photo of a PLA plastic base with a winding of HTS tapes. 1 – winding made of HTS-tapes, 2 – base for HTS winding.

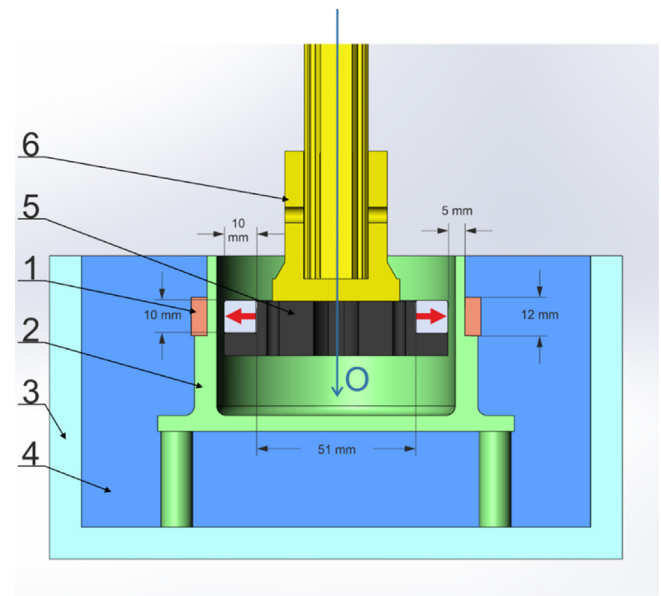


Fig. 14. Diagram of the experimental setup for torque measurement in a radial coupler. 1 – winding made of HTS-tapes, 2 – holder of HTS tapes, 3 – cryostat, 4 – liquid nitrogen, 5 – driving half-coupler with PMGs, 6 – beam connected to a torque sensor. Red arrows show the magnetization direction of the PMGs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

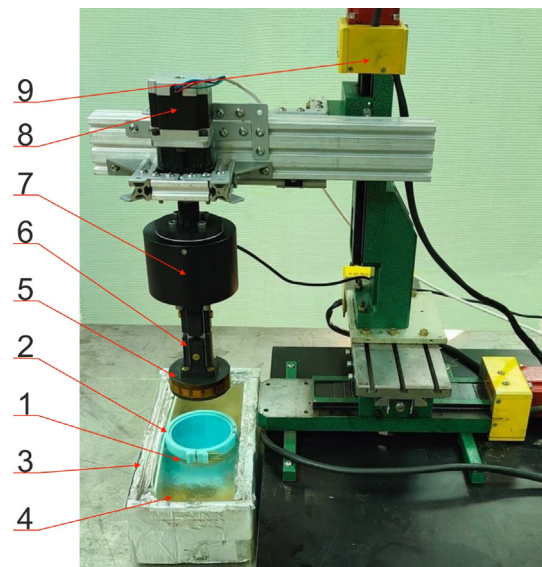


Fig. 15. Photo of the experimental setup for torque measurement in a radial coupler. 1 – winding made of HTS-tapes, 2 – holder of HTS tapes, 3 – cryostat, 4 – liquid nitrogen, 5 – driving half-coupler with PMGs, 6 – beam connected to a torque sensor, 7 – a torque sensor, 8 – stepper motor, 9 – positioning system. Cooling in ZFC mode is shown.

tom, liquid nitrogen does not flow into the base and does not interact with the driving half-coupler. The coupler half is mounted on a beam (6 in Figs. 14 and 15), connected to a static torque sensor (7 in Fig. 15). A 3-component force sensor MC 15-3B-200 manufactured by Dacell was used as a torque sensor. The sensor, in turn, is connected to the rotation system – stepper motor (8 in Fig. 15). The stepper motor is mounted on a 3-coordinate positioning system (9 in Fig. 15).

The HTS-tape was used exactly the same as in the preliminary studies of adhesion in HTS couplers of various configurations. The tape was

wound in 10 and 20 turns. We chose 20 turns because, as mentioned earlier, in our experience, for tapes with a critical current of 550A when interacting with most commercially available PMGs [16], the optimal ratio of levitation force to the number of layers in a stack is 20 layers. To study the influence of the number of turns in the system on the torque, it was decided to carry out measurements on windings with a different number of turns, namely 10 turns.

The measurements were carried out in two modes – in the field cooling mode (FC) and in the zero field cooling mode (ZFC). In FC mode, the system was cooled in the measurement position shown in Fig. 14. In ZFC mode, the superconductor was cooled with the driving half of the coupler (and thus permanent magnets) removed far from the HTS. Thus, the superconductor is cooled in a zero external magnetic field. After this, the driving coupler half was lowered to the measurement position shown in Fig. 14. The superconductor enters the field of permanent magnets of the driving coupler's half. Thus, in both cooling modes, measurements take place in the same position shown in Fig. 14, only the superconductor cooling histories differ.

3.2. Results and discussion

The results of measurements of the torque dependence on the rotation angle for a winding of 10 turns and 20 turns are shown in Fig. 16 and Fig. 17, respectively. Let us consider the dependence in ZFC mode for windings of 10 and 20 turns. The dependence is a monotonically increasing function reaching saturation. The dependence increases sharply for angles up to 10 degrees.

This monotonic dependence is due to the fact that all Abrikosov vortices that entered the superconductor under the influence of permanent magnets of the coupler half move following the magnetic field of the coupler half. Due to the hard pinning of vortices in a superconductor, the remnant field begins to lag relative to the magnetic field. This delay leads to the emergence of a torque that tends to return system to original position. At the same time, the steady-state torque for windings of 10 turns (0.215 N·m) is more than twice as high as for windings of 20 turns (0.1 N·m). This is due to the fact that the magnetic field penetrates to a greater depth for windings of 10 tapes, a larger area is captured by vortices. Consequently, a large area is remagnetized when the coupler half rotates.

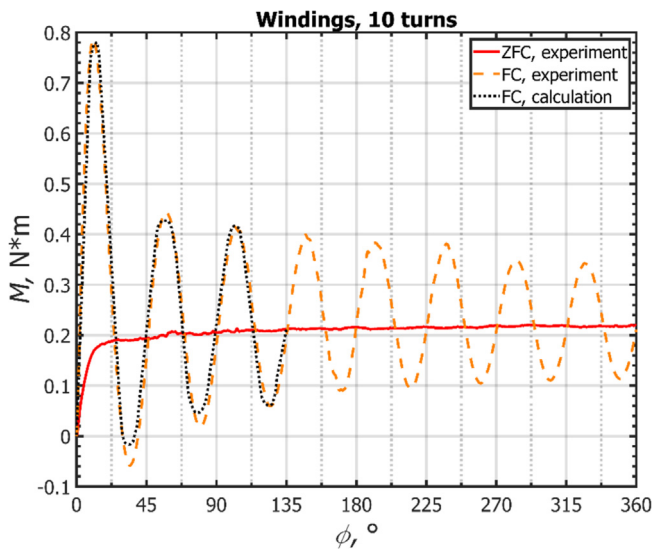


Fig. 16. Dependence of the torque between the coupler halves on the angle of rotation for a radial coupler based on 10-turn HTS-windings. Results are presented for FC and ZFC modes. Liquid nitrogen was used for cooling. For the FC mode, the results of numerical calculations are also presented.

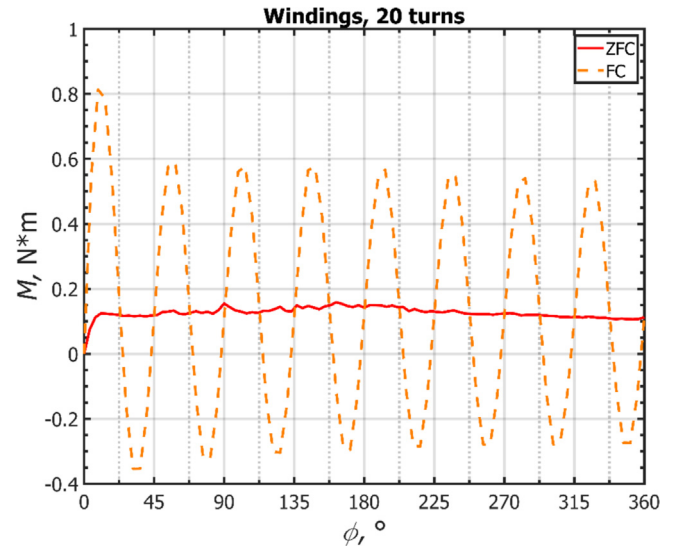


Fig. 17. Dependence of the torque between the coupler halves on the angle of rotation for a radial coupler based on 20-turn HTS-windings. Results are presented for FC and ZFC modes. Liquid nitrogen was used for cooling.

The torque dependence on the rotation angle for the FC case is similar to the dependences obtained on stacks of HTS-tapes. They represent a periodic dependence with a period of 45 degrees. The moment of force in the first cycle also exceeds its values in subsequent cycles. For windings of 10 tapes, a finite element calculation was carried out using the model we described earlier. The calculation results are in good agreement with the experimental data, which is another confirmation of the correctness of the developed model.

The maximum torque value for windings of different thicknesses is practically the same and is equal to a value of about 0.8 N·m. It should also be noted that the maximum torque on one stack of 50 tapes was 0.025 N·m. Based on this, one would expect a value of 0.4 N·m on 16 stacks of 50 tapes. However, we obtained a value twice as large on windings of only 20 tapes. This may indicate greater efficiency in the use of windings compared to stacks of HTS tapes.

Similar to the results obtained for individual stacks of HTS-tapes, the average torque value is shifted to the positive region. From both dependences it is clear that the average value coincides with good accuracy with the torque in ZFC mode. For windings of 20 turns, the average torque value significantly exceeds the oscillation amplitude, similar to what was observed for stacks of tapes (Fig. 5 and Fig. 7). While for windings of 10 turns, the average value of the moment of force is shifted relative to zero by a significant value, exceeding the oscillation amplitude.

This behavior of the system is associated with the peculiarities of the dynamics of currents and magnetic flux in the thickness of the superconducting winding for the FC and ZFC modes during magnetization reversal. To demonstrate this, Figs. 18 and 19 shows the dynamics of the change in the normal component of the current during rotation of the permanent magnets. It is clearly seen that in the ZFC mode, the currents induced in the winding by the external magnetic field follow the magnetic field of the PMGs. At the same time, the FC mode is characterized by the processes of displacing the magnetic flux into the thickness of the winding during magnetization reversal. In turn, this can lead to the fact that part of the magnetic flux will follow the external magnetic field, and part of the magnetic flux in the thickness will participate in the screening of the magnetic field of the PMGs and the magnetized outer layers. It is also worth noting that since HTS winding of 10 turns at a temperature of 77.4 K is initially more uniformly magnetized, the portion of currents that go into the sample during magnetization reversal is significantly smaller. These differences in the

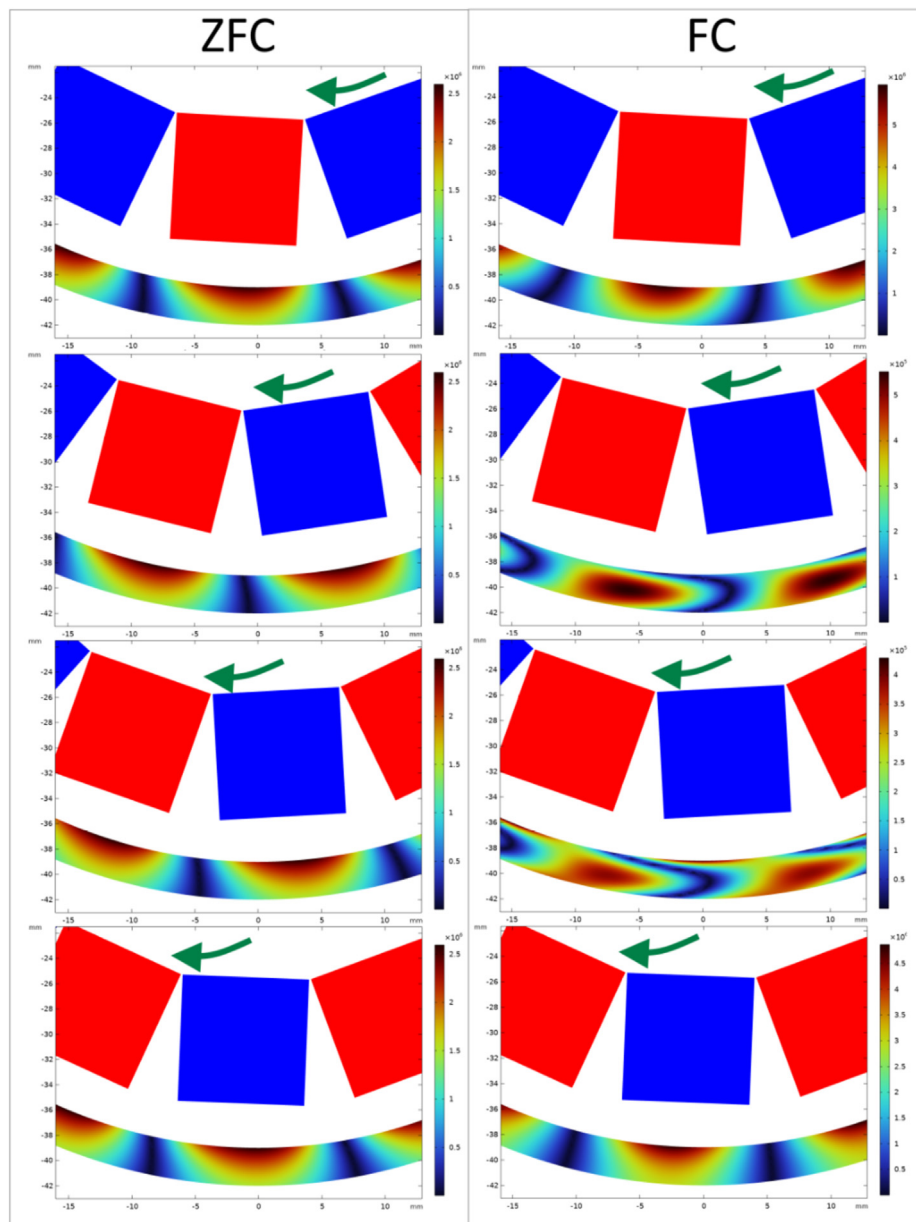


Fig. 18. Distributions of the normal current component in a superconducting winding of 20 turns under magnetization in the ZFC and FC modes. Uniform blue and red colors illustrate the position of the PMGs with oppositely directed magnetization vectors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

behavior of currents and magnetic flux determine differences in the average value of the torque for HTS couplings. In practice, this leads to a slightly better ability to resist magnetization reversal for a coupling with a thicker winding. In addition, other current redistribution processes that cannot be taken into account within the framework of a two-dimensional numerical model will also affect the value of the torque.

The point is that the phenomenon of a shift in the average torque value can be explained as follows. In FC case, HTS windings capture a significantly larger magnetic flux. The configuration of the vortices is established, aligned along the magnetic field coinciding with the magnetic field of the coupler half made of PMG magnets. When rotated from the equilibrium position, areas of the HTS-tapes closer to the edges will begin to remagnetize due to the alternating field from the PMGs. Similar to ZFC case, the vortices will begin to move after the field with a delay, leading to the appearance of a constant torque.

These moving vortices will fill the edge regions of the HTS winding. In the inner part of the tape, the magnetization remains almost unchanged; the vortices are rigidly attached to the defects, acting as permanent magnets. The magnetizing reversal region of the windings near the edges of the tape will produce a constant contribution similar to the ZFC mode, while the interior of the tape will produce a periodic contribution similar to permanent magnets (similar to what is presented in Fig. 4). The only difference between the periodic contribution and permanent magnets will be that the magnetic moments of the current domains in the central part of the HTS will decrease due to the creep of the magnetic flux. The vortices will leave the central region, annihilating with opposite vortices at the boundary of the region of the moving magnetic flow. This leads to a gradual attenuation of the captured magnetic flux, and, as a consequence, to a decrease in the amplitude of torque oscillations. The dependences of the oscillation amplitude on the number of full rotations of the coupler

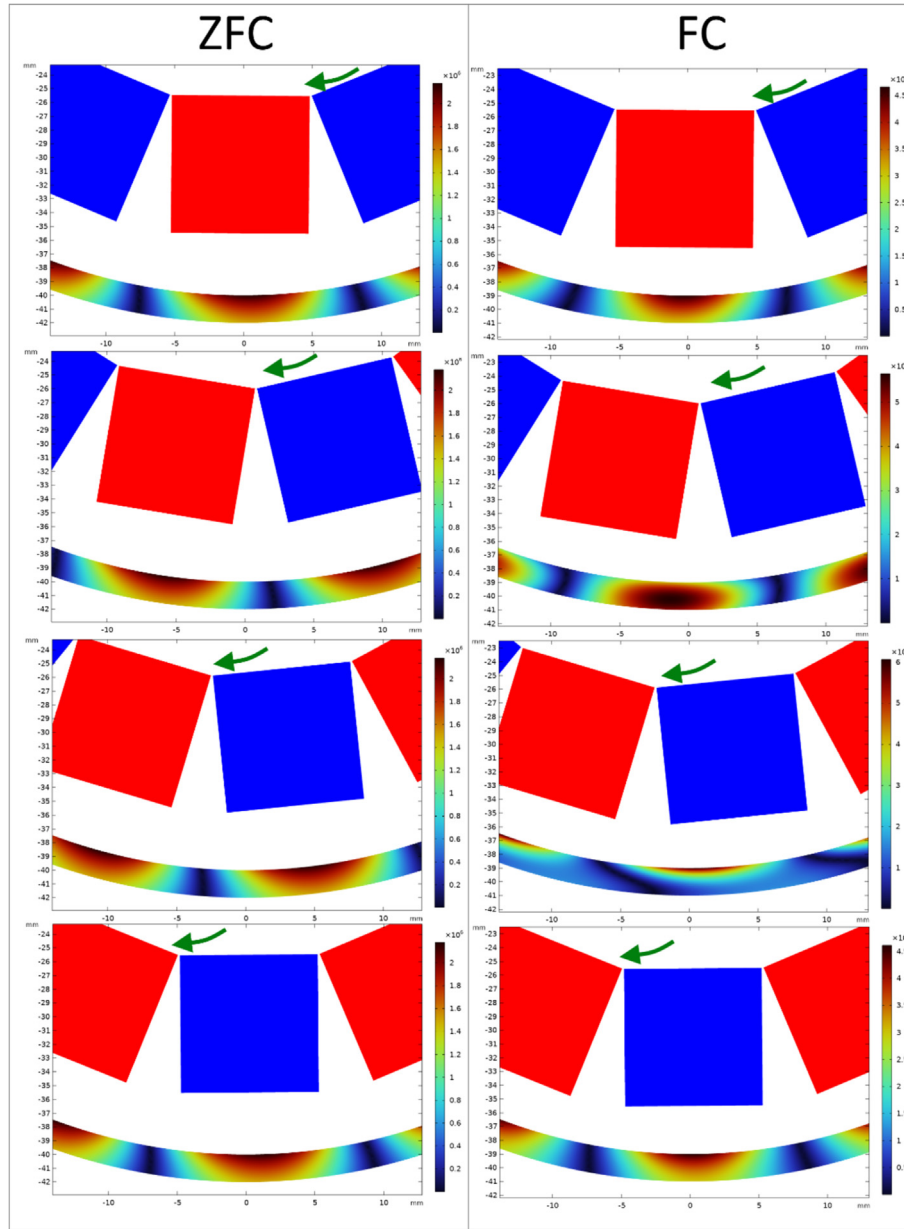


Fig. 19. Distributions of the normal current component in a superconducting winding of 10 turns under magnetization in the ZFC and FC modes. Uniform blue and red colors illustrate the position of the PMGs with oppositely directed magnetization vectors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

half are shown in Fig. 20 for 10 full rotations. After 10 full rotations, the decay of the amplitude of the moments of forces slows down significantly. Predictions based on approximation of experimental data show that both dependences reach saturation relatively quickly. After approximately 150 revolutions, an amplitude change of less than one percent is expected. The final steady-state torques for 10 and 20 turns in the windings are $0.22 \pm 0.15 \text{ N}\cdot\text{m}$ and $0.1 \pm 0.39 \text{ N}\cdot\text{m}$.

Thus, in terms of maximum torque, windings of 20 turns do not provide a noticeable advantage. In the mode of slipping of the coupler halves, the average torque will be equal to the moment in the ZFC mode. In this case, the average torque of windings of 10 tapes also exceeds windings of 20 turns. It is also more efficient to use windings of 10 turns if one plan to use the coupler in ZFC mode. The only case when 20-turn windings are superior to 10-turn windings is non-slip rotation after partial slip, where the torque between the coupler halves is determined by the sum of the average force moment and the torque

amplitude. The resulting maximum torque values are $0.37 \text{ N}\cdot\text{m}$ for 10 turns windings and $0.49 \text{ N}\cdot\text{m}$ for 20 turns windings. Thus, windings of 20 turns give an advantage of 30 % in this case.

4. Conclusions

Experimental studies of the rotational load characteristics of magnetic couplers were carried out. Two types of magnetic couplers were considered: magnetic couplers based on HTS elements (HTS-PMG couplers) and based on permanent magnets (PMG-PMG couplers). Radial and axial coupler configurations are considered, with coupler halves consisting of 16 elements. Based on the experimental data obtained, it is shown that the torque in the radial superconducting couplers exceed the torque in the axial superconducting couplers. Differences in the torques are especially noticeable at large distances between the coupler halves. Also, in the dependence of the torque on the angle

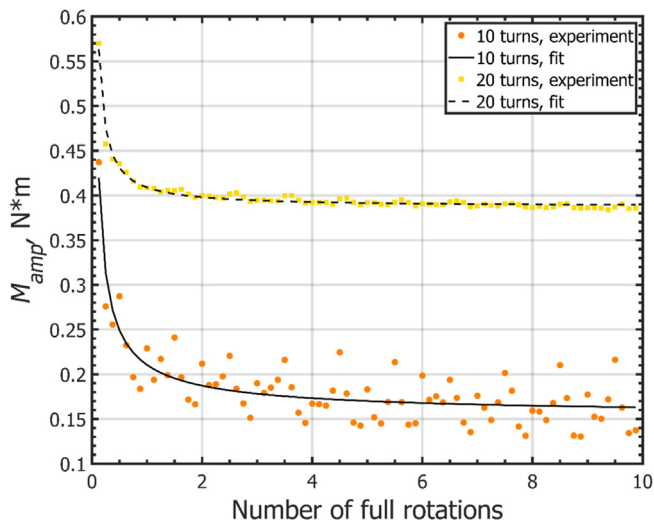


Fig. 20. Dependence of the oscillation amplitude of the torque between the coupler halves on the number of full rotations for HTS windings of 10 and 20 turns.

of rotation, areas of instability are observed for the case of an axial superconducting coupler. These facts make the axial type of HTS couplers less preferable for use. However, this only applies to the configurations we have considered. For a more complete comparison of the two types of couplers, it is necessary to optimize each type of coupler and compare the optimized designs, which is beyond the scope of this work.

For the case of stacks of 50 tapes cooled with liquid nitrogen, HTS-PMG couplers are 10 times inferior to PMG-PMG couplers. To optimize the characteristics of HTS couplers, a numerical model was developed and the power characteristics of PMG and HTS couplers were simulated. The modeling was performed using the finite element method using a mixed A-T-H formulation. To verify the model, the forces of levitation interaction between the coupler elements were calculated for configurations similar to the experimental ones. The calculated data are in good agreement with the experimental data in the entire range of rotation angles for PMG couplers. For the case of HTS-PMG couplers, the calculated curves begin to diverge from the experimental ones at rotation angles of more than 25 degrees, when a significant remagnetization of HTS-tapes occurs. However, the model predicts the maximum torque in coupler quite accurately.

The model was used to optimize HTS-PMG couplers. The initially proposed radial HTS coupler configuration was modified by adding HTS windings, lowering the operating temperature and using stronger permanent magnets. The results of modeling the load characteristics of an optimized HTS-PMG coupler indicate that a magnetic drive based on HTS-tapes can achieve characteristics similar to traditional PMG-PMG coupler. Also, the simulation results show that HTS-couplers are superior to PMG-PMG couplers in damping external disturbances.

Due to the fact that the radial coupler showed the best results in terms of torque, a model of this type of coupler was assembled based on HTS-tapes windings and a study of its rotational load characteristics was carried out. Couplers with windings of 10 and 20 turns were studied. It is shown that the couplers can be used in both FC and ZFC modes. In ZFC mode, the torque dependence on the angle of rotation reaches a constant, and in the FC mode, periodic oscillations are additionally added to this constant.

In ZFC mode, as well as in FC mode when the coupler halves slip, the 10 turn assembly shows twice the maximum torque. In FC mode, if slippage stops, the 20 turn windings show a 30 % advantage in maxi-

mum adhesive torque. In the absence of slippage in FC mode, windings of 20 tapes do not show a noticeable advantage. Thus, increasing the number of turns of HTS windings in radial HTS-PMG couplers does not always lead to an improvement in the rotational load characteristics of the coupler.

CRediT authorship contribution statement

Maxim Osipov: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Irina Martirosian:** Visualization, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Aleksandr Starikovskii:** Validation, Investigation. **Dmitrii Aleksandrov:** Visualization, Investigation. **Sergei Pokrovskii:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [Maxim Osipov, Irina Martirosyan, Aleksandr Starikovskii, Dmitrii Aleksandrov, Sergei Pokrovskii reports financial support was provided by Ministry of Science and Higher Education of the Russian Federation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

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